

RynnBrain 1.1 System Card

 <https://alibaba-damo-academy.github.io/RynnBrain.github.io>
 <https://github.com/alibaba-damo-academy/RynnBrain>
 <https://huggingface.co/collections/Alibaba-DAMO-Academy/rynnbrain>
 https://www.modelscope.cn/collections/DAMO_Academy/RynnBrain

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1 Introduction

Embodied intelligence requires models to go beyond visual recognition and language understanding. A robot operating in the physical world must perceive objects [49, 47], reason about spatial relations [52, 69], identify actionable regions [22], and translate such understanding into executable behaviors [5, 4]. While recent multimodal large language models have achieved strong performance on image, video, and visual question answering tasks [37, 66, 23, 27], real-world robotic systems impose additional requirements: they must understand three-dimensional structure, reason across viewpoints, ground language instructions into physical locations, and support downstream manipulation policies. Therefore, the central challenge for embodied foundation models is not only whether they can understand visual content, but whether such understanding is grounded in the physical world and, crucially, whether it can be transferred to real-robot action [54].

Recent embodied foundation models [42, 55] have advanced rapidly along different axes, including object and region localization, spatio-temporal understanding, perspective-taking, physical reasoning, and robot-oriented planning. RynnBrain 1.0 [17] explored this direction by introducing a unified embodied multimodal model for egocentric understanding [27, 15, 43], spatial grounding, physically grounded reasoning, and planning, demonstrating the feasibility of combining general multimodal capabilities with robot-oriented spatial supervision. However, several key aspects were not fully studied in the previous version. First, RynnBrain 1.0 did not systematically investigate whether embodied capabilities can scale under a unified training recipe. Second, although it learned rich spatial grounding abilities, large-scale 3D data [25] had not yet been introduced as a central component of the training mixture. Third, its robot-oriented interface and real-robot validation—in particular, how well the base model supports downstream VLA post-training—were still preliminary. These limitations motivate RynnBrain 1.1: a model family that studies embodied scaling across different model sizes, integrates large-scale 3D-grounded data into the spatio-temporal embodied brain, and improves the interface between foundation-model grounding and downstream manipulation.

To address these limitations, we present RynnBrain 1.1, a systematic upgrade of the RynnBrain model family. Built upon Qwen3.5 [44], RynnBrain 1.1 includes three model scales: 2B, 9B, and 122B-A10B. Scaling the embodied foundation model to the 122B-A10B level is a central contribution of this release. All models follow a unified capability definition and training framework, allowing us to study how embodied perception, spatio-temporal understanding, spatial reasoning, grounding, and planning evolve with model scale.

In addition to model scaling, RynnBrain 1.1 introduces explicit 3D-grounded training into the embodied foundation model. Real robots operate in three-dimensional environments [57, 25], where understanding object depth, relative spatial relations, object pose, and actionable regions is essential for reliable interaction. We therefore extend the existing training mixture with large-scale 3D data from both real-world and synthetic sources [45, 18, 8]. These data complement the 2D and video-based supervision used in previous training by providing more direct signals about physical-space structure. Through this design, RynnBrain 1.1 aims to strengthen not only 3D grounding itself, but also broader spatio-temporal understanding and robot-oriented perception, such as contact point prediction [56, 22].

We evaluate RynnBrain 1.1 from two complementary perspectives. First, we benchmark the base models

on a broad suite of embodied multimodal tasks covering spatio-temporal understanding, spatial reasoning, localization, and native 3D grounding. The results show an overall empirical scaling trend from 2B to 9B and 122B-A10B, while the 2B and 9B models demonstrate strong language-conditioned 3D grounding after incorporating explicit 3D supervision.

Second, we evaluate how embodied pretraining transfers to downstream VLA learning. Under the same post-training recipe and demonstration data, we compare RynnBrain-VLA with a Qwen-based VLA and representative generalist VLA baselines. To support training across heterogeneous robot morphologies and control interfaces, we introduce a unified action space that organizes embodiment-specific actions into semantically aligned body-part groups, while using embodiment-specific masks to activate only the dimensions available on each robot. This design allows data from different embodiments to be jointly trained within a single policy without forcing incompatible low-level action spaces to match.

RynnBrain-VLA is deployed on three heterogeneous platforms, a Unitree G1 humanoid, an Astribot bimanual robot, and a Tianji-Wuji dexterous-hand system, through a unified cross-embodiment framework. Controlled evaluations on Astribot and Tianji-Wuji show that RynnBrain provides a stronger initialization than the general-purpose Qwen base model. Joint multi-task and multi-embodiment training further improves the average process score and final success rate over separately fine-tuned per-task policies.

Overall, RynnBrain 1.1 makes the following contributions:

- We release RynnBrain 1.1, a new embodied foundation model family spanning 2B, 9B, and 122B-A10B scales. The models share a unified capability definition and training framework, enabling an empirical study of embodied capability scaling from compact dense models to a large sparse-MoE model.
- We introduce explicit 3D-grounded training into the 2B and 9B models, enabling native language-conditioned 3D grounding with competitive performance against specialized detectors and proprietary systems. This extends RynnBrain from image-plane localization toward physical-space perception that can support downstream robotic applications.
- We develop a robust RynnBrain-VLA with a unified cross-embodiment action space and embodiment-specific masking. The policy is deployed on a humanoid robot, a bimanual manipulator, and a dexterous-hand system, and demonstrates strong real-world performance across heterogeneous tasks and control interfaces. Unified Action Space and multi-embodiment training further improves over separately fine-tuned per-task policies.

2 Overview

2.1 Model Architecture

An overview of the RynnBrain architecture is shown in [Figure 1](#).

RynnBrain 1.1 adopts a decoder-only vision–language architecture following the design principles of Qwen3.5 [44]. It consists of a vision encoder, a vision–language projector, and a large language model (LLM) backbone initialized from Qwen3.5 variants. In this release, we build three model scales, namely RynnBrain-1.1-2B, RynnBrain-1.1-9B, and RynnBrain-1.1-122B-A10B. All models share the same overall architecture and training formulation, enabling a systematic study of embodied scaling across model sizes. We also adopt DeepStack [38] and Interleaved MRoPE [24] to better integrate multimodal information and support long-context spatio-temporal modeling over images, videos, and embodied observations.

2.2 Training Infrastructure

As described in RynnBrain [16], our training pipeline is designed for large-scale embodied multimodal data with highly heterogeneous sequence lengths. We follow the same infrastructure design, including online sequence-length-aware load balancing, per-sample loss reduction, memory-efficient training with ZeRO and gradient checkpointing, and optimized MoE execution with expert parallelism and efficient token dispatching. These components allow us to train on mixed image, video, text, and spatial-grounding data while mitigating straggler effects caused by long-tail sequence-length distributions.

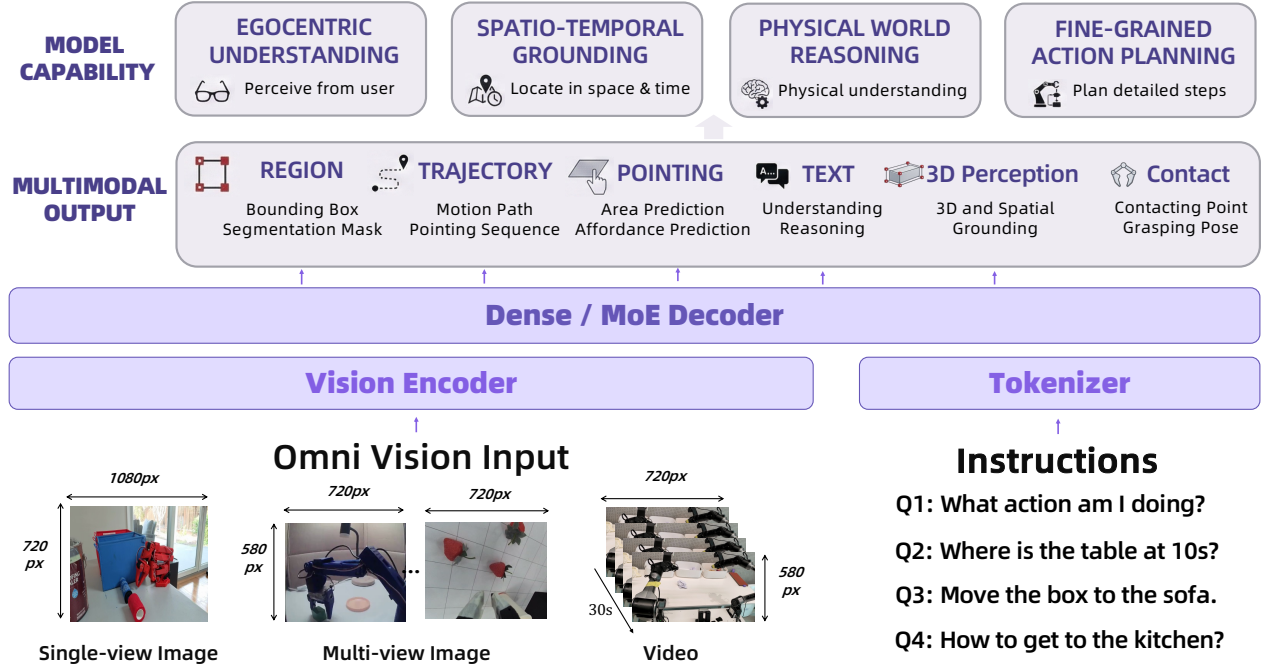


Figure 1 Overview of the RynnBrain architecture. RynnBrain processes omni vision inputs, including single view images, multi view images, and videos, together with language instructions. A shared dense or mixture of experts decoder generates aligned multimodal outputs, including text, regions, trajectories, pointing, 3D Perception and Contacting signals. This unified output space supports egocentric understanding, spatiotemporal grounding, physically grounded reasoning, and fine grained action planning in real world environments.

For RynnBrain 1.1, we extend this infrastructure to the new Qwen3.5-based model family, covering 2B, 9B, and 122B scales. The same training pipeline is used across model sizes to ensure that differences in downstream performance mainly reflect model scaling and data updates rather than changes in infrastructure. For the 122B-A10B model, we adopt sparse MoE training with expert parallelism and communication-efficient token dispatching, following the infrastructure principles established in RynnBrain [16].

3 Physics-Aware Spatio-temporal Pretraining

Enabling generalizable robots to interact naturally with real-world environments requires a foundation model that can perceive over time, ground its predictions in physical space, and scale across different computational budgets. Following RynnBrain [16], we retain two core design principles for embodied pretraining: spatio-temporal memory and physical-world grounding. Spatio-temporal memory allows the model to build representations over historical visual observations, including objects, locations, events, and motion patterns. Physical-world grounding requires the model to express part of its understanding through explicit spatial outputs, such as objects, regions, affordances, and trajectories, rather than purely free-form language. RynnBrain 1.1 extends this recipe in two directions. First, we instantiate the same spatio-temporal embodied pretraining framework at three model scales, 2B, 9B, and 122B-A10B, enabling us to study how embodied perception, reasoning, localization, and planning evolve with model size. Second, as an initial exploration of native 3D grounding, we incorporate large-scale 3D-grounded data into the pretraining mixtures of RynnBrain 1.1-2B and RynnBrain 1.1-9B. This setting allows us to evaluate how explicit 3D supervision contributes to physical-space understanding across different model scales.

3.1 Training Recipe

To equip RynnBrain with spatio-temporal memory and physical grounding, we adopt a unified pretraining framework that maps multimodal inputs into a shared semantic representation space. The training recipe is structured around two core components: a unified input-output representation and a physics-aware

optimization strategy.

Unified Spatio-temporal Representation. Following RynnBrain [16], RynnBrain 1.1 treats images and videos as a unified visual modality. Formally, a visual input $\mathbf{V}$ is represented as a sequence of frames $\{I_t\}_{t=1}^T$, where $T = 1$ for static images and $T > 1$ for videos. For videos, frames are uniformly sampled to preserve temporal continuity. Each frame is encoded into visual tokens and augmented with temporal positional embeddings to encode frame order. This representation enables RynnBrain 1.1 to capture temporal dependencies, motion patterns, and trajectory dynamics across extended visual sequences.

Physically Grounded Output Space. To ensure physical-world grounding, we explicitly structure the output space to bridge high-level reasoning and spatial prediction. Unlike conventional vision-language models that treat spatial quantities as free-form text, RynnBrain 1.1 represents spatial outputs with discrete coordinate tokens. Spatial entities such as bounding boxes $\mathcal{B}$, points $\mathcal{P}$, and trajectory waypoints $\mathcal{T}$ are normalized to the range $[0, 1000]$ and encoded as integer tokens. This discretization converts continuous spatial prediction into a classification problem, allowing the model to generate precise and physically meaningful spatial outputs using the same autoregressive mechanism as language generation. For the main scaling study, we keep this output formulation consistent across the 2B, 9B, and 122B-A10B models, so that performance differences mainly reflect model scale and data mixture changes.

In addition, we incorporate explicit 3D annotations into the pretraining mixtures of RynnBrain 1.1-2B and RynnBrain 1.1-9B to explore native 3D grounding. These models follow the same instruction-following and autoregressive training paradigm as the rest of the RynnBrain 1.1 family, while extending the supervision from image-plane grounding toward physical-space understanding.

Specifically, the model takes as input a natural language instruction and camera intrinsics, and is trained to directly predict a 3D bounding box parameterized in the camera coordinate system—including center position (cx, cy, cz) , box dimensions (w, l, h) , and orientation $(pitch, yaw, roll)$, all in physical units (meters and radians). This 9-dimensional output format follows the design of prior works such as Seed-VL [50] and Qwen3-VL [3], which also adopt a compact set of numerical parameters for spatially grounded predictions. To preserve the unified autoregressive framework, we discretize all continuous 3D quantities into integer tokens within a fixed range, analogous to our 2D spatial tokenization. This design enables the model to perform physically grounded reasoning from a single image, inferring not only what and where in the pixel plane, but also how objects are positioned and oriented in real-world 3D space, directly bridging visual perception and downstream robotic or navigation tasks.

Optimization. RynnBrain 1.1 is trained end-to-end using a standard next-token prediction objective. The training loss is defined as:

$$\mathcal{L} = - \sum_{i=1}^L \log P(y_i | y_{<i}, \mathbf{V}, \Theta), \quad (1)$$

where $\mathbf{V}$ denotes the visual input, $\mathbf{y}$ is the mixed sequence of textual and coordinate tokens, and Θ represents the model parameters. We use the same objective for all RynnBrain 1.1 variants, including the 3D-grounded 2B and 9B preview models. Optimization hyperparameters are adjusted for each model scale based on pilot experiments conducted on representative subsets of the pretraining data. Detailed training configurations are reported in Table 1.

Table 1 Hyperparameters of the pretraining stage for the RynnBrain model series.

Parameter	RynnBrain1.1-2B	RynnBrain1.1-9B	RynnBrain1.1-122B-A10B
Base Model	Qwen3.5-2B	Qwen3.5-9B	Qwen3.5-122B-A10B
Optimizer	AdamW	AdamW	AdamW
Learning Rate	5×10^{-6}	2×10^{-6}	2×10^{-6}
Learning Rate Vision	1×10^{-6}	2×10^{-6}	2×10^{-6}
Global Batch Size	512	1024	1024
Warmup Ratio	0.03	0.03	0.03

Table 2 Pretraining data mixture for RynnBrain 1.1. New or substantially expanded data sources compared with RynnBrain 1.0 are highlighted in red.

Category	Sub-Task	Data Sources
General MLLM	General	LLaVA-OV-SI [31], LLaVA-Video [67], ShareGPT-4o-video [10], VideoGPT-plus [35], FineVideo [20], CinePile [48], ActivityNet [6], YouCook2 [70], LLaVA-SFT [34], VL3 [66] , RynnBrain-Thinking
Cognition	Object Understanding	RynnBrain-Object, RefCOCO [62], Google Refexp [36], Osprey-724K [64], DAM [33], VideoRefer-700k [65]
	Spatial Understanding	SenseNova-SI-800K [8], VSI-590K [59], VLM-3R [18], RynnBrain-Spatial
	Counting	RynnBrain-Counting, Molmo2 [12]
	OCR	RynnBrain-OCR, Llama-Nemotron-VLM OCR [40]
	Egocentric Task Understanding	EgoRe-5M [43], EgoTaskQA [27], Env-QA [1], QAEgo4D [21], RoboVQA [51], Robo2VLM [9], ShareRobot [26]
Localization	Object Localization	ADE20K [68], COCOStuff [7], Mapillary [39], PACO-LVIS [46], PASCAL-Part [11], VG [29], RoboAfford-Object [22], RynnBrain-Grounding
	Area Localization	RefSpatial [69], RoboAfford-Area [22], Molmo2 [12], RynnBrain-Area, RoboRefer [69] ,
	Affordance Localization	RynnBrain-Affordance, RoboAfford-Affordance [22]
	Trajectory Prediction	RynnBrain-Trajectory, FSD [63], MolmoAct [30]
	3D Grounding	WildDet3D Essential [25] , WildDet3D Synthetic [25] , FoundationPose [57]
Planning	Grasp / Contact Point Prediction	Grasp-Anything-Visible [56] , GraspClutter6D [2] , GraspNet-1B [19] , Jacquard V2 [32] , GraspFactory [53]
	Manipulation	AgibotWorld [14], Open X-Embodiment [13], RynnBrain-Planning, Galaxea-G0 [28]

3.2 Pretraining Data

RynnBrain 1.1 follows the capability-oriented data organization of RynnBrain [16]. As shown in Table 2, we organize the pretraining mixture into general multimodal data, cognition data, spatio-temporal localization data, 3D-grounded data, robot-oriented perception data, and planning data. For data types already introduced in RynnBrain 1.0, we follow the same instruction-following format and data processing protocol. Specifically, visual inputs are represented as images or videos, and model outputs are converted into mixed sequences of text and spatial tokens, including boxes, points, trajectories, and grasp-related annotations. In this section, we focus on the data updates introduced in RynnBrain 1.1.

General MLLM Data. To preserve broad multimodal understanding, we continue to use a general-purpose image-video-text corpus covering image understanding, video understanding, visual question answering, temporal reasoning, and instruction following. The mixture includes commonly used MLLM datasets such as LLaVA-OV-SI [31], LLaVA-Video [67], ShareGPT-4o-video [10], VideoGPT-plus [35], FineVideo [20], CinePile [48], ActivityNet [6], YouCook2 [70], and LLaVA-SFT [34]. In RynnBrain 1.1, we further incorporate self-collected RynnBrain-Thinking data. RynnBrain-Thinking is included as part of the general mixture to better align the model with the Qwen3.5-Thinking style. All data are converted into the unified instruction-following format before being mixed for pretraining.

Multi-dimensional Cognition Data. For embodied cognition, RynnBrain 1.1 follows the task taxonomy of RynnBrain [16], covering object understanding, spatial understanding, counting, OCR, and egocentric task understanding. Object understanding data include RynnBrain-Object, RefCOCO [62], Google Refexp [36], Osprey-724K [64], DAM [33], and VideoRefer-700K [65], which provide object-centric supervision for fine-grained recognition and reasoning. Spatial understanding data include SenseNova-SI-800K [8], VSI-590K [59],

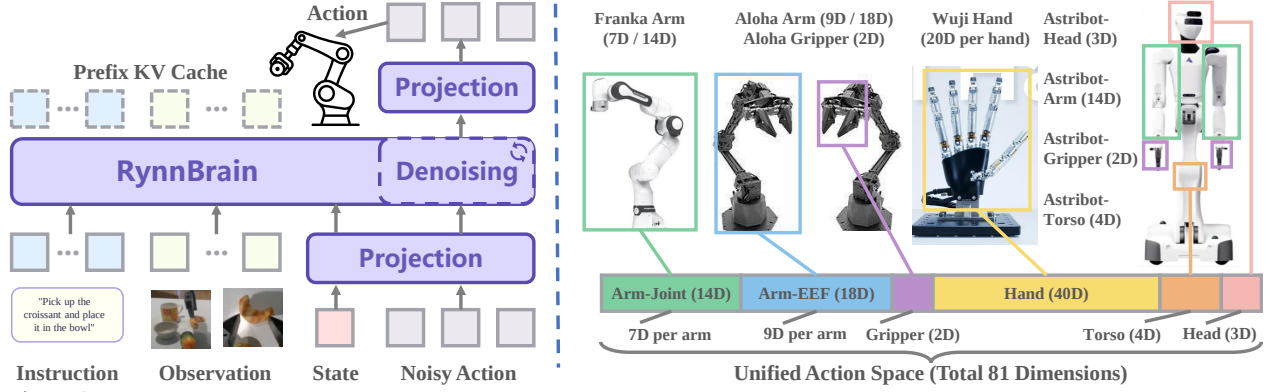
VLM-3R [18], and RynnBrain-Spatial, supporting reasoning over spatial relations, viewpoints, and scene structure. For counting and OCR, we use RynnBrain-Counting, Molmo2 [12], RynnBrain-OCR, and the newly added Llama-Nemotron-VLM OCR data. For egocentric task understanding, we use EgoRe-5M [43], EgoTaskQA [27], Env-QA [1], QAEgo4D [21], RoboVQA [51], Robo2VLM [9], and ShareRobot [26]. These datasets are converted following the RynnBrain instruction format to provide broad supervision for object-centric, spatial, textual, and first-person task understanding.

Spatio-temporal Localization Data. RynnBrain 1.1 inherits the localization formulation of RynnBrain 1.0, where spatial outputs are represented using normalized coordinate tokens. The localization mixture covers object localization, area localization, affordance localization, trajectory prediction, and grasp/contact-related prediction. Object localization uses ADE20K [68], COCOStuff [7], Mapillary [39], PACO-LVIS [46], PASCAL-Part [11], VG [29], RoboAfford-Object [22], and RynnBrain-Grounding. Area localization uses RefSpatial [69], RoboAfford-Area [22], Molmo2 [12], RynnBrain-Area, and the newly added RoboRefer. Affordance localization uses RynnBrain-Affordance and RoboAfford-Affordance [22]. Trajectory prediction uses RynnBrain-Trajectory, FSD [63], and the newly added MolmoAct. For these data sources, we follow the RynnBrain processing protocol and convert different annotations into a shared output space consisting of boxes, points, and trajectories.

3D-grounded Data. To explore native 3D grounding, we conduct a 3D-grounded preview training in the 2B and 9B setting by mixing explicit 3D annotations into the pretraining data. Given an image, a natural language instruction, and camera intrinsics, the model is trained to predict a 3D bounding box in the camera coordinate system. The box is parameterized by its center position (cx, cy, cz) , dimensions (w, l, h) , and orientation (pitch, yaw, roll), represented in physical units such as meters and radians.

To support this at scale, we construct a multi-source corpus combining large-scale automatically lifted annotations with high-fidelity synthetic data. For broad category and scene coverage, we adopt **WildDet3D-Data** [25], a recently proposed open-vocabulary 3D detection dataset that lifts 2D annotations from COCO, LVIS, Objects365, and V3Det into 3D via a multi-model candidate generation and selection pipeline. We use its two curated subsets after additional filtering: *Essential* (102,979 images, 374K manually verified annotations after filtering out geometrically implausible candidates and small objects, 12,064 categories) and *Synthetic* (896,004 images, 888K automatically selected annotations, 11,896 categories, filtered by a fine-tuned Molmo2 scoring six perceptual criteria with a retention threshold >10), offering unprecedented category diversity but inheriting residual noise from monocular lifting. For each 3D annotation, we leverage Qwen-VL to generate an open-ended referring query based on the corresponding 2D bounding box and category label, incorporating spatial descriptions (e.g., relative positions such as "left," "right," "front," or "behind") and attribute modifiers (e.g., color, size, or material cues) to make the referring expression more discriminative and precise for grounding. To complement this with physically exact supervision, we additionally incorporate **FoundationPose** [57], a synthetic dataset derived from the Google Scanned Objects subset under controlled camera trajectories, which provides 1,849K images with noise-free 3D bounding boxes computed directly from ground-truth object-to-camera transformations, with each asset assigned a natural-language category via Qwen-VL to bridge to real-world semantics. Though its category coverage is limited to 928 instances, its metric-scale accuracy and zero-depth-ambiguity geometry serve as a strong regularizer for depth and orientation prediction, teaching correct inductive biases for camera-to-world projection when mixed with the diverse but noisier WildDet3D annotations.

Contacting point Data. RynnBrain 1.0 represented planar grasps using the four corners of an oriented rectangle. We found this detector-style representation poorly aligned with action grounding in a general embodied model. A target object typically admits a set of functionally valid grasps rather than a unique rectangle. Moreover, unlike an object bounding box, a grasp rectangle has no canonical image-space extent: its width, height, aspect ratio, and boundary placement depend on gripper geometry and dataset-specific annotation conventions. Consequently, IoU against a single reference rectangle may penalize functionally equivalent grasps and is not necessarily correlated with execution success. The four-corner representation also introduces redundant extent variables and geometric consistency constraints into autoregressive generation. RynnBrain 1.1 therefore adopts a compact contact-centered representation ($a = (p, \theta)$), where $(p = (x, y))$ denotes the center of the annotated grasp configuration and (θ) is the in-plane angle of the line connecting the gripper fingers. The coordinates are normalized to $[0, 1000]$, and each target is serialized as `<grasp pose> (x, y),  $\theta$` . This representation factors out annotation- and embodiment-dependent rectangle extents



while preserving an action-relevant spatial anchor and orientation cue.

A substantial portion of our source data consists of simulation-native, multidimensional grasp annotations rather than directly usable image-text pairs. We therefore develop a view-conditioned rendering and projection pipeline to convert these annotations into reliable 2D supervision. For each object-grasp configuration, we select a camera pose subject to multiple observability constraints: the target object must remain fully inside the image, occupy an appropriate range of image scales, and avoid severe truncation or occlusion; meanwhile, the projected grasp center and orientation must both remain visible and spatially distinguishable. Views in which the object is excessively near, distant, truncated, or visually ambiguous, or in which the grasp configuration is not observable, are discarded. The retained views are rendered as RGB images, after which the multidimensional grasp annotations are projected into the common $((p, \theta))$ representation. This procedure is essential because unconstrained camera sampling can produce geometrically valid annotations that cannot be inferred from the rendered image.

After rendering, projection, and filtering, the resulting corpus contains 2.6M training samples from Grasp-Anything [56], Jacquard V2 [32], GraspFactory [53], GraspNet-1B [19], and GraspClutter6D [2]. We treat $((p, \theta))$ as an action-grounding interface rather than a complete executable grasp pose; depth, approach direction, gripper aperture, collision constraints, and robot kinematics are resolved by the downstream manipulation policy.

Physics-aware Planning Data. For manipulation planning, RynnBrain 1.1 follows the structured planning format of RynnBrain [16], where high-level task instructions are paired with grounded sub-task descriptions involving objects, areas, and affordances. The planning mixture includes AgibotWorld [14], Open X-Embodiment [13], RynnBrain-Planning, and the newly added Galaxea-G0. For existing grounded planning data, we follow the same annotation format and conversion protocol as RynnBrain 1.0. For Galaxea-G0, the original annotations provide sub-task-level labels but do not contain explicit high-level task instructions. We therefore construct the corresponding global task descriptions using Gemini 3.1 Pro, conditioned on the annotation IDs and the sequence of annotated sub-tasks. The resulting data are then converted into the same grounded planning format as the rest of the planning mixture.

4 Post-training for Vision-Language-Action Model

We use downstream VLA post-training as both a practical capability and a probe of base-model quality. This section describes the VLA *method*: the policy architecture (Section 4.1), the unified action space design (Section 4.2), the cross-embodiment / cross-driver deployment framework (Section 4.3), Real-Time Chunking for smooth low-latency control (Section 4.4), and the fine-tuning setup shared by all backbones (Section 4.5). The corresponding real-robot results are reported in Section 5.6.

4.1 Model Architecture

To bridge the gap between planning and physical execution, we propose **RynnBrain-VLA**, which translates fine-grained plans into executable robot actions. We build RynnBrain-VLA upon the RynnBrain base model to utilize the large-scale pretraining on fine-grained object references and precise spatial localization while maintaining low inference latency. The overall model architecture is shown in Figure 2. Generally, we adopt a flow matching framework to predict an action chunk [4] at each step. The VLM backbone is served as a single-stream Diffusion Transformer (DiT) taking a single packed sequence containing the condition and the noisy actions as input. To make the VLM compatible with this framework, we add three linear projections to align the dimension of the input noises, input timestamp embeddings, and the output actions to the VLM hidden size. To preserve the VLM’s inherent instruction-following capabilities, we utilize its native conversation format for organizing the input sequence. Consistent with the planning model, pointing information is passed in a text-based format, with the task’s initial frame prepended to the input sequence.

```
<|im_start|>user
INSTRUCTION:
<start_frame>
Pick the <affordance> (x,y) </affordance> of the <object> (x0,y0),(x1,y1) </object>
OBSERVATION:
<camera_1><camera_2><camera_3>
STATE:
<state>
What action should the robot take ?<|im_end|>
<|im_start|>assistant
<action>
```

Following π_0 [4], actions are positioned at the end of the sequence to enable the KV cache during inference.

4.2 Unified Cross-Embodiment Action Space

As illustrated in Figure 2, RynnBrain-VLA represents actions from heterogeneous robot embodiments within a shared high-dimensional action space. Before training, we partition the unified action vector into semantically aligned groups corresponding to different body parts and control modes, including left and right arms, parallel grippers, dexterous hands, torso, head, and mobile-base control. Each embodiment activates only the action groups that are physically available on that robot, while all remaining dimensions are suppressed by an embodiment-specific mask.

Each embodiment activates only the action groups that are physically available on that robot, while all remaining dimensions are suppressed by an embodiment-specific mask. The action loss is computed only over the active dimensions. For the Unitree G1 humanoid, the unified prediction is mapped to the SONIC whole-body controller. Tianji-Wuji activates the bilateral arm and dexterous-hand groups while masking torso and mobile-base dimensions. Atribot activates the bilateral arm, parallel-gripper, torso, head, and mobile-base groups, while masking the high-dimensional dexterous-hand joints.

This formulation allows trajectories from different robot morphologies to be mixed within the same training batch and optimized by a single VLA policy. Shared action components, such as left- and right-arm motion, receive supervision from multiple embodiments, whereas embodiment-specific components are updated only by the corresponding data. Consequently, common manipulation structures—including reaching, grasping, bimanual coordination, and task-level interaction patterns—can be transferred across robots without forcing incompatible low-level action spaces to match. The unified action space therefore serves not only as a common output interface, but also as a mechanism for cross-embodiment knowledge sharing while preserving embodiment-specific control.

4.3 Cross-Embodiment and Cross-Driver Deployment

Real-world deployment requires a single policy interface to drive robots with very different morphologies and low-level controllers. We therefore design a cross-embodiment, cross-driver framework that decouples the

VLA policy from embodiment-specific execution, enabling the *same* RynnBrain-VLA recipe to be adapted to heterogeneous robots. The framework has three components:

- **Unified action interface.** Predicted action chunks are expressed in a normalized action representation that is mapped to each embodiment’s native action space (*e.g.* end-effector pose, joint targets, or dexterous-hand joint angles). **[TODO: specify action dimensionality per embodiment and normalization scheme]**
- **Observation adapter.** Heterogeneous camera layouts and proprioceptive state are converted into the fixed multi-camera + state input format expected by the policy. **[TODO: camera count / resolution / state definition per embodiment]**
- **Driver abstraction.** A thin driver layer maps normalized actions to each robot’s low-level controller, so that adding a new embodiment or controller does not change the policy. **[TODO: describe drivers for Unitree G1, Astribot, Tianji-Wuji]**

We instantiate this framework on three heterogeneous embodiments—the Unitree G1 humanoid, the Astribot bimanual manipulator, and the Tianji-Wuji dexterous hand—covering distinct action spaces, degrees of freedom, and end-effectors. **[TODO: control frequency and any embodiment-specific handling]**

4.4 Real-Time Chunking (RTC)

Executing action chunks in a naive, strictly sequential manner introduces pauses at chunk boundaries and couples the effective control frequency to the model’s inference latency, resulting in visibly non-smooth motion. To address this, we adopt **Real-Time Chunking (RTC)**, which overlaps the inference of the next action chunk with the execution of the current one and blends overlapping actions across chunk boundaries. **[TODO: describe RTC mechanism precisely: chunk length H , overlap window, blending / consistency scheme]** RTC yields smoother trajectories and a stable control frequency under on-robot inference. We quantify its effect on both action smoothness and latency in ??.

4.5 Fine-tuning Setup

We collect teleoperation demonstrations on the three target embodiments described above, covering a range of pick-and-place and manipulation tasks with different objects and interaction requirements. Each demonstration contains synchronized visual observations, robot states, and action trajectories, together with a task instruction specifying the target object or placement location.

For the controlled comparison between the Qwen-based VLA and RynnBrain-VLA in [Section 5.6](#), we use the same demonstration data, VLA post-training pipeline, and optimization settings. Both policies are fine-tuned for 60k steps with a learning rate of 2×10^{-5} and a batch size of 32. All input images are proportionally resized such that the short side is 384 pixels. This setup reduces differences introduced by downstream training and allows us to examine how the choice of base model affects the resulting real-robot policy. The fine-tuned policies are evaluated quantitatively in [Section 5.6](#).

5 Evaluation

5.1 Embodied Cognition Capability

We evaluate RynnBrain 1.1 on a diverse suite of embodied cognition benchmarks covering video-based spatial intelligence, multi-view reasoning, embodied question answering, robot-oriented spatial understanding, and object-centric cognition. The evaluation includes VSI-Bench [58], MMSI [60], ERQA [54], RoboSpatial-VQA [52], MindCube [61], EmbSpatial, RynnBrain-Object, and RynnBrain-Spatial. As shown in [Tables 3 and 4](#), RynnBrain 1.1 improves substantially over the previous RynnBrain models on reasoning-intensive and spatially demanding tasks. At the 2B scale, the improvements are task-dependent: MMSI increases from 34.1 to 40.5, MindCube from 50.1 to 61.7, and RynnBrain-Spatial from 57.2 to 63.8, while several

Table 3 Comparison with lightweight (2B-scale) models on embodied cognition and localization benchmarks. * denotes results from our reproduction.

Benchmark	RynnBrain 1.1 2B	RynnBrain 2B	Cosmos 3 -Edge 2B	SenseNova-SI-1.1 InternVL3 2B	Cosmos -Reason2 2B	Qwen3.5 2B
VSI-Bench	72.9	70.5	59.2	63.7*	46.3*	44.1
MMSI	40.5	34.1	32.3	34.2	27.8*	28.3
ERQA	42.3	42.3	42.0	28.7*	37.8	33.0
RoboSpatial-Pointing	59.6	65.6	-	0.0*	11.5*	-
RoboSpatial-VQA	60.4	64.9	-	24.8*	65.3*	-
RoboSpatial-Overall	62.1	65.2	-	16.1*	47.4*	41.5
MindCube	61.7	50.1	-	41.8	33.7*	37.5
EmbSpatial	73.9	79.6	-	63.0*	-	67.7*
RynnBrain-Object	74.6	70.7	24.0	20.7	25.8	37.1
RynnBrain-Spatial	63.8	57.2	22.6	13.0	24.3	15.6
RefSpatial-Bench	58.5	52.7	48.4	5.6*	23.6*	30.0
RynnBrain-Grounding	84.1	79.1	51.6	12.6	14.5	24.2
RynnBrain-Area	57.2	54.6	39.1	8.6	22.1	8.1
RynnBrain-Affordance	90.2	89.4	77.6	57.8	55.5	80.4
RynnBrain-Trajectory	68.8	66.6	58.7	49.0	43.7	28.3

RoboSpatial and EmbSpatial results remain below RynnBrain-2B. The gains become more systematic at the 9B scale, where RynnBrain 1.1 outperforms RynnBrain-8B on eight of ten shared benchmarks. In particular, MindCube improves from 56.6 to 86.9, MMSI from 39.6 to 47.0, and RynnBrain-Spatial from 59.9 to 67.9, indicating stronger multi-view and spatial reasoning. Compared with other models at similar scales, RynnBrain 1.1 achieves leading results on multiple cognition benchmarks at both the 2B and 9B scales. RynnBrain 1.1-122B-A10B further extends this advantage to the large-model regime, with performance increasing from 40.5 to 47.0 and 52.0 on MMSI, and from 61.7 to 86.9 and 89.6 on MindCube, when scaling from 2B to 9B and 122B-A10B. These results show that model scaling is particularly beneficial for reasoning-intensive embodied cognition.

5.2 Embodied Location Capability

We evaluate embodied location across six spatial grounding tasks: RefSpatial-Bench [69], RoboSpatial-Pointing [52], RynnBrain-Grounding, RynnBrain-Area, RynnBrain-Affordance, and RynnBrain-Trajectory. These benchmarks cover language-guided spatial reference, object and region localization, affordance grounding, and trajectory prediction. RynnBrain 1.1 improves over the previous RynnBrain models on all six benchmarks at both the 2B and 9B scales. At 2B, RefSpatial-Bench increases from 52.7 to 58.5 and RynnBrain-Grounding from 79.1 to 84.1. At 9B, the gains become larger on relational and motion-oriented tasks: RefSpatial-Bench improves from 59.2 to 67.2, while RynnBrain-Trajectory increases from 64.5 to 70.0. The relatively small gain on RynnBrain-Affordance reflects its already high performance in RynnBrain 1.0. Both RynnBrain 1.1-2B and RynnBrain 1.1-9B achieve the best results among models in their respective scale groups on most of these location benchmarks. At the large scale, RynnBrain 1.1-122B-A10B achieves leading performance on RefSpatial-Bench, RynnBrain-Grounding, and RynnBrain-Affordance. The clearest scaling trend appears on RefSpatial-Bench, which improves from 58.5 at 2B to 67.2 at 9B and 79.1 at 122B-A10B, showing that relational spatial grounding benefits strongly from increased model capacity.

5.3 Scaling Analysis

A central contribution of RynnBrain 1.1 is studying how embodied capability evolves with model scale under a unified training recipe. We compare RynnBrain 1.1 (2B, 9B, 122B-A10B) against the corresponding raw Qwen3.5 baselines at matched model sizes, grouping benchmarks into three categories: *general embodied*

Table 4 Comparison with 4B–9B scale models on embodied cognition and localization benchmarks. * denotes results from our reproduction.

Benchmark	RynnBrain 1.1	RynnBrain	Cosmos 3 -Nano	RoboBrain-2.5	Thinker	Molmo2-ER	Pelican-VL	MiMo- Embodied	Cosmos -Reason2	SenseNova -SI1.5	Qwen3.5
	9B	8B	8B	4B	4B	5B	7B	7B	8B	8B	9B
VSI-Bench	74.9	70.95	54.9	41.7	65.4	74.5	52.8	48.5	53.7*	67.3*	61.0
MMSI	47.0	39.6	36.2	20.5	27.7*	43.8	26.2*	30.2*	31.3*	38.3*	13.4
ERQA	47.5	46.8	46.0	43.3	42.5*	46.8	39.8*	46.8	46.0*	40.0*	41.5*
RoboSpatial-Pointing	69.7	60.1	33.0*	-	-	32.0	-	-	24.6	-	48.7
RoboSpatial-VQA	68.0	80.0	77.6*	-	-	73.4	-	-	76.3	61.4*	57.0
RoboSpatial-Overall	69.1	73.1	62.1*	62.3	70.8	59.0*	57.5	61.8	59.0*	40.1*	54.1*
MindCube	86.9	56.6	40.5*	26.9	39.0*	57.0	33.7*	43.1*	43.8*	92.1	33.1
EmbSpatial	81.9	80.4	81.4*	76.9	80.2	78.8	76.6	76.2	77.0*	80.3	83.0
RynnBrain-Object	75.9	71.2	39.8	36.4	26.5	17.1	30.8	39.0	37.2	16.8	45.6
RynnBrain-Spatial	67.9	59.9	26.1	32.6	17.3	19.7	20.5	28.3	31.4	19.9	28.4
RefSpatial-Bench	67.2	59.2	53.1	56.0	61.0	52.5	22.3	48.0	33.1*	8.6*	37.3
RynnBrain-Grounding	85.4	81.6	72.9	5.4	2.4	4.2	3.5	49.8	60.0	44.7	33.0
RynnBrain-Area	59.6	56.2	52.0	42.4	33.8	30.6	46.5	49.4	37.6	11.3	32.9
RynnBrain-Affordance	90.7	90.4	85.1	69.1	59.9	81.7	81.4	84.4	83.9	64.9	60.5
RynnBrain-Trajectory	70.0	64.5	67.9	63.0	49.2	59.3	59.2	61.3	64.0	53.9	45.0

cognition (visual recognition, spatial understanding, and object-centric reasoning), *reasoning-intensive cognition* (multi-view spatial reasoning and video-based temporal inference), and *embodied localization* (language-guided spatial grounding and trajectory prediction). Figure 3 summarizes the scaling trends across these groups.

Embodied capabilities exhibit non-uniform scaling laws. Unlike standard vision–language benchmarks where scaling consistently improves performance, we observe three distinct scaling regimes in the embodied domain. (1) For *general embodied cognition*, both RynnBrain 1.1 and Qwen3.5 improve monotonically with scale, and the performance gap steadily narrows. This mirrors conventional MLLM scaling—larger models reliably improve on recognition and comprehension tasks, and general model scaling can partially compensate for the lack of embodied supervision. (2) For *reasoning-intensive cognition*, the two models diverge fundamentally: RynnBrain 1.1 improves steadily (+38.6%), while Qwen3.5 exhibits *negative scaling* (−39.2%), with the gap widening from 18.2 to 50.8. This indicates that multi-view and temporal reasoning capabilities are not emergent from general VLM scaling but require explicit embodied supervision as a prerequisite. Without spatiotemporal grounding, stronger language priors in larger models may override weak visual-spatial signals, producing more confident but less accurate responses. (3) For *embodied localization*, both models improve substantially, yet Qwen3.5 at its largest scale still underperforms RynnBrain 1.1 at its smallest. This suggests that for spatial-temporal pointing tasks, *data scaling*—training on explicit coordinate supervision—is more effective than model scaling alone. RynnBrain’s embodied pretraining equips even its compact model with structured spatial output capabilities that pure language modeling must recover implicitly, requiring disproportionately more parameters to approximate.

Embodied pretraining as a scaling enabler. The decomposition in Figure 3(d) further illustrates this contrast. RynnBrain 1.1 consistently establishes a higher performance floor across all categories, reflecting the value of embodied data even at compact scales. More critically, on reasoning-intensive tasks, embodied pretraining determines whether scaling *helps or hurts*—it transforms a capability that degrades under pure VLM scaling into one that benefits consistently. These findings indicate that for the embodied domain, data-driven spatial supervision and model scaling are complementary rather than substitutable: scaling amplifies the benefit of embodied pretraining on reasoning-heavy tasks, while embodied supervision ensures that scaling produces reliable gains rather than regression.

Table 5 Comparison with large-scale open-source and proprietary models on embodied cognition and localization benchmarks. * denotes results from our reproduction.

Benchmark	RynnBrain 1.1 122B-A10B	Cosmos 3 Super 32B	RoboBrain 2.0 32B	Cosmos- Reason2 32B	Pelican-VL 72B	Qwen3.5 122B-A10B	Gemini 3 Pro -	GPT 5.4 -	Claude Sonnet 4.6 -	Gemini Robotics -ER 1.5 -	HY-Embodied 0.5 A32B
VSI-Bench	75.0	60.9	42.7	67.0*	57.3	66.6*	48.8*	49.2*	44.4*	45.8	68.3
MMSI	52.0	41.8	28.5*	32.1*	30.7*	9.2*	49.2	37.5*	33.0*	-	39.2
ERQA	54.3	51.2	46.0	45.3	43.0	62.0	70.5	47.5*	44.8*	54.8	62.3
RoboSpatial-Pointing	69.7	24.2*	-	34.4	-	59.8	27.0*	20.8*	5.8*	31.1	-
RoboSpatial-VQA	70.6	74.1*	-	67.11	-	68.9	66.7*	75.0*	55.3*	79.3	-
RoboSpatial-Overall	70.3	56.7*	72.4	55.4	55.4	65.7*	53.0*	56.1*	38.0*	62.6	-
MindCube	89.6	40.2*	29.2	39.4*	32.5*	30.8*	70.8	10.4*	21.0*	54.7	69.2
EmbSpatial	83.7	83.8*	-	78.5*	76.6	80.5	83.6	81.8*	61.5*	78.4	-
RynnBrain-Object	77.0	48.3	26.2	36.8*	42.2	55.4	58.4	55.5	33.5	-	-
RynnBrain-Spatial	70.0	34.5	11.6	38.4*	32.2	31.0	38.5	43.3	36.7	-	-
RefSpatial-Bench	79.1	57.0	54.0	45.5*	49.5	69.3	65.5	26.7*	14.6*	48.5	57.2
RynnBrain-Grounding	86.6	74.4	0.0	60.0	10.8	66.6	62.1	24.2	3.3	-	-
RynnBrain-Area	60.7	53.0	45.3	37.6	53.2	41.9	61.5	37.1	15.8	-	-
RynnBrain-Affordance	91.3	84.6	76.1	83.9	87.3	79.0	86.0	83.4	60.0	-	-
RynnBrain-Trajectory	69.7	69.3	60.3	64.0	64.1	62.8	72.0	72.8	56.7	-	-

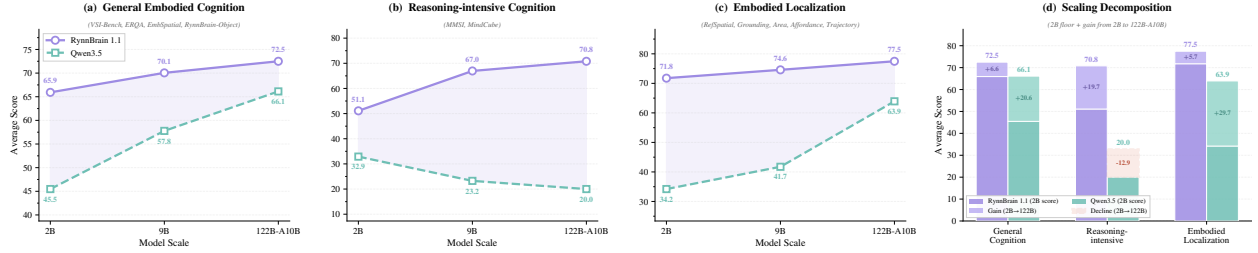


Figure 3 Scaling analysis of RynnBrain 1.1 vs. Qwen3.5 from 2B to 122B-A10B. (a)–(c) Average scores across three capability groups: general embodied cognition, reasoning-intensive cognition, and embodied localization. (d) Scaling decomposition showing the 2B performance floor and the gain from scaling to 122B-A10B.

5.4 3D Grounding

We evaluate the native 3D grounding capability of RynnBrain 1.1 at both the 2B and 9B scales. On SUN RGB-D, RynnBrain 1.1-2B achieves 34.28 AP@15, already outperforming general-purpose VLMs such as Seed1.5-VL (33.5) and Gemini 2.0 Pro (32.5). Scaling the model to 9B further improves the result to **41.12 AP@15**, substantially narrowing the gap to the closed-source Gemini Robotics-ER (48.3). These results show that explicit 3D-grounded training can equip even compact models with metric spatial understanding, while larger model capacity further strengthens depth, size, and orientation reasoning.

A similar scaling trend is observed on WildDet3D-Bench. RynnBrain 1.1-2B obtains 17.36 AP3D, while RynnBrain 1.1-9B reaches **23.44 AP3D**, surpassing the specialized WildDet3D detector trained with additional in-domain data (22.6). The improvement from 2B to 9B indicates that native 3D grounding benefits from model scaling in addition to explicit 3D supervision. Overall, RynnBrain 1.1 demonstrates strong language-conditioned 3D grounding across model sizes, with the 9B model achieving competitive performance against both specialized detectors and substantially larger proprietary systems.

5.5 Contact Point Prediction

Evaluating contact point prediction remains challenging because there is currently no standardized metric that reliably reflects the functional validity of a predicted contact. Pixel-level distance to a single annotation may penalize alternative contact points that are equally valid for manipulation, while conventional grasp-box metrics are not directly applicable to our contact-centered representation. We therefore provide a qualitative evaluation in Figure 5 to illustrate the capability of RynnBrain 1.1 across diverse objects and instructions.

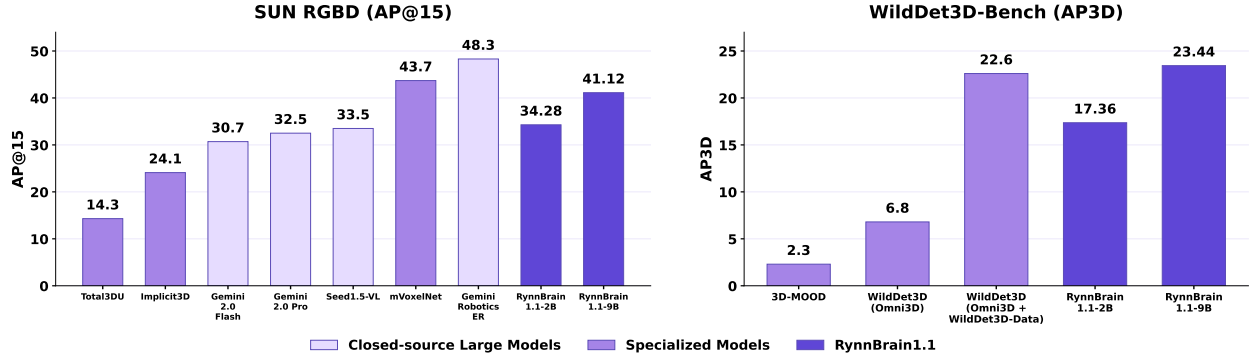


Figure 4 Performance Comparison on SUN RGB-D and WildDet3D-Bench with RynnBrain1.1-2B and 9B.

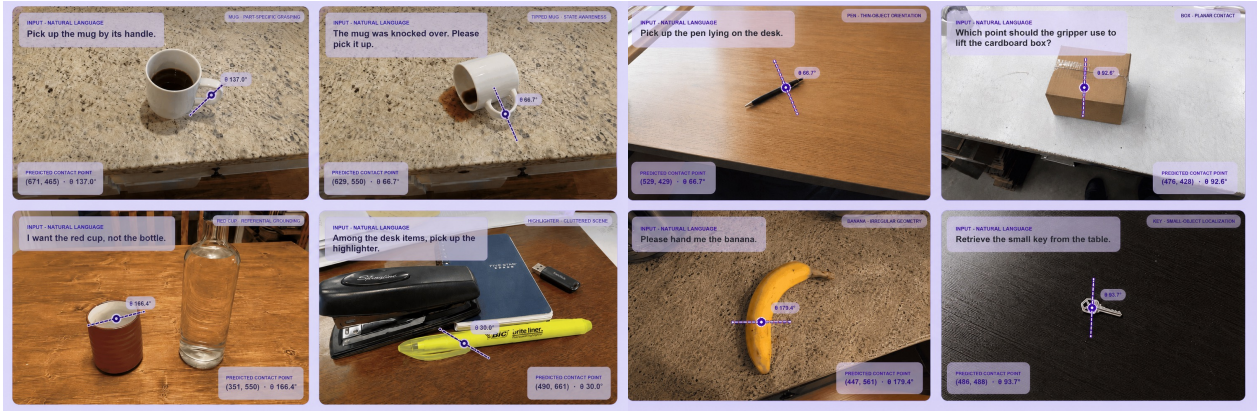


Figure 5 Qualitative results of contact point prediction by RynnBrain 1.1. Given an image and a natural-language instruction, the model predicts an action-relevant contact point together with its in-plane grasp orientation. The examples cover diverse object geometries, object states, target parts, and cluttered scenes, demonstrating instruction-conditioned contact grounding beyond object-level localization. Coordinates are normalized to $[0, 1000]$.

Here we show the cases inferred by RynnBrain1.1-9B.

The visualized cases show that RynnBrain 1.1 can identify task-relevant contact locations rather than merely predicting object centers. For example, the model selects the handle when asked to pick up a mug, adapts its prediction when the mug is overturned, and produces appropriate contact locations and orientations for thin objects, boxes, and irregularly shaped items. It also distinguishes the instructed target in cluttered or referential settings. These results demonstrate that RynnBrain 1.1 can jointly reason about object identity, geometry, state, and intended interaction when generating contact-oriented spatial predictions.

5.6 Real-Robot Evaluation

We evaluate RynnBrain-VLA on real robots to answer three questions: (i) can RynnBrain-VLA achieve stable performance on real-world manipulation tasks? (ii) does a stronger embodied foundation model lead to a better VLA policy? and (iii) can joint training on data from heterogeneous robot embodiments provide mutual benefits across embodiments? We further report the effect of RTC and discuss practical deployment limitations.

5.6.1 Experimental Setup

Evaluation Protocol. We deploy RynnBrain-VLA on three heterogeneous embodiments, as summarized in Table 6 and Figure 6. For each task, we conduct 20 trials with randomized initial object placements and report the final task success rate under a fixed success criterion. Since some long-horizon tasks can be naturally decomposed into multiple sub-tasks, final success alone does not fully reflect the execution progress of a policy.

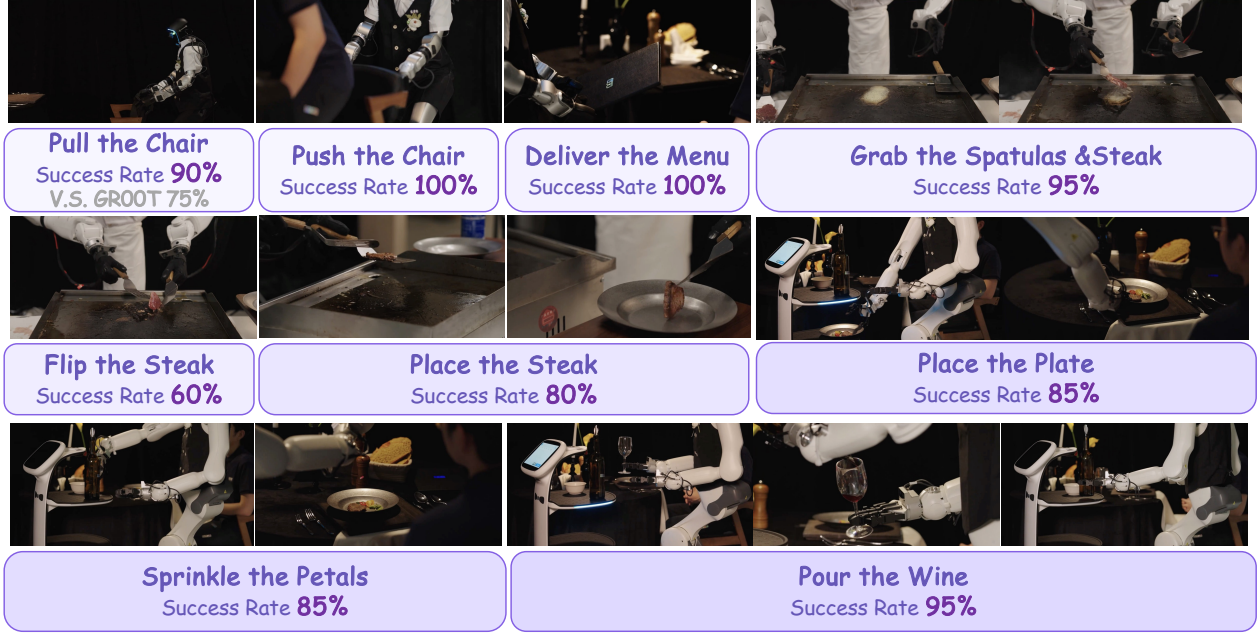


Figure 6 Real-robot evaluation across all tasks considered in our experiments. The tasks are performed in diverse real-life environments and cover whole-body manipulation, bimanual coordination, dexterous-hand operation, object placement, furniture interaction, and object delivery. Representative execution snapshots and success rates over 20 trials are shown for each task.

We therefore additionally report a process score that measures the fraction of completed sub-tasks. For a task with K sub-tasks, the process score is defined as

$$\text{ProcessScore} = \frac{1}{20K} \sum_{i=1}^{20} \sum_{j=1}^K s_{i,j}, \quad (2)$$

where $s_{i,j} \in \{0, 1\}$ indicates whether the j -th sub-task is successfully completed in the i -th trial. This metric provides a more fine-grained measure of task progress, especially for long-horizon manipulation tasks in which a policy may complete several intermediate stages without reaching the final goal.

Deployment and Action Space. For all tasks, inference is performed locally on a workstation equipped with a single NVIDIA GeForce RTX 4090 GPU. The workstation is directly connected to the robot and transmits the predicted actions for real-time execution. For tasks involving the Unitree G1 humanoid, we employ SONIC as the whole-body controller. The model predicts a 64-dimensional universal motion token together with two 7-dimensional tokens representing the joint configurations of the left and right arms, respectively. These tokens are concatenated into a 78-dimensional action representation and decoded by the SONIC decoder into target positions for the actuated joints of the Unitree G1.

Baselines and Training Settings. We select GR00T N1.7 [41] and $\pi 0.5$ [5] as representative generalist VLA baselines. We additionally construct Qwen-VLA by applying the same VLA post-training recipe used for RynnBrain-VLA to the Qwen base model. For RynnBrain-VLA, we consider two training settings. In the single-task setting, a separate policy is fine-tuned for each individual task. In the generalist setting, we jointly fine-tune a single policy on multi-task and multi-embodiment data using our unified action space, and denote the resulting model as RynnBrain-VLA_{Generalist}. To obtain the best deployment performance for each policy, we use method-specific action-chunk lengths rather than enforcing a shared chunk size across all models. Specifically, we use an action-chunk length of 40 for GR00T N1.7, 50 for $\pi 0.5$, and 32 for RynnBrain-VLA, including both the single-task and generalist settings, and Qwen-Based-VLA.

Tasks. We design three challenging long-horizon manipulation tasks for quantitative comparison: *Sprinkle the Petals*, *Pour the Wine*, and *Grab the Spatulas*. Each task consists of multiple temporally ordered sub-tasks and requires the policy to complete a sequence of perception, reaching, grasping, manipulation, and

Table 6 Real-robot evaluation results. Each task reports the process score and final success rate (%). The process score measures the average fraction of completed sub-tasks, while a trial is counted as successful only when all sub-tasks of the corresponding task are completed.

Model	Sprinkle the Petals		Pour the Wine		Grab the Spatulas		Average	
	<i>Astribot</i>		<i>Astribot</i>		<i>Tianji-Wuji</i>		–	
	Proc.	Succ.	Proc.	Succ.	Proc.	Succ.	Proc.	Succ.
Qwen-Based-VLA	75.00	65.00	75.00	65.00	55.00	50.00	68.33	60.00
GR00T-1.7	86.25	75.00	77.00	65.00	86.67	80.00	83.31	73.33
$\pi_{0.5}$	65.00	60.00	69.00	65.00	83.33	70.00	72.44	65.00
RynnBrain-VLA	87.50	85.00	88.00	80.00	98.33	95.00	90.94	85.00
RynnBrain-VLA _{Generalist}	90.56	85.00	97.00	95.00	96.67	95.00	94.74	91.67

placement operations. The task setups are illustrated in Figure 6. The first two tasks are performed on the Astribot robot with gripper-based whole-body control, where the robot needs to actively change its viewpoint during execution. In *Sprinkle the Petals*, the policy must continuously observe the container and determine whether the petals have been fully poured out before proceeding, requiring closed-loop visual observation and task-progress understanding. *Pour the Wine* requires the robot to precisely align the bottle opening with the target glass and execute a controlled pouring motion, placing stronger demands on fine-grained affordance understanding and precise manipulation. *Grab the Spatulas* is performed with Tianji-Wuji dexterous hands and requires fine-grained grasping and hand control to reliably manipulate the tools. Together, these tasks cover complementary challenges in long-horizon reasoning, visual feedback, precise affordance-guided manipulation, and dexterous interaction, and serve as the main benchmark for comparing different VLA methods and training settings. Beyond the three main tasks, we further evaluate RynnBrain-VLA on a series of additional robot tasks across heterogeneous embodiments, including object placement, flipping, furniture interaction, and object delivery. These evaluations broaden the coverage of embodiments and manipulation primitives and provide an additional assessment of policy generality.

5.6.2 Stable Real-World Performance.

As shown in Figure 6 and Table 6, RynnBrain-VLA achieves stable performance across multiple robot embodiments, including the Astribot bimanual platform, the Tianji-Wuji dexterous-hand system, and the Unitree G1 humanoid. The policy performs reliably under substantially different action spaces and control interfaces, covering gripper-based whole-body manipulation, dexterous grasping, bimanual coordination, and humanoid whole-body control. In particular, the Astribot tasks require the robot to actively change its viewpoint during execution and continuously update its actions from visual feedback. The strong results on these tasks indicate that RynnBrain-VLA remains robust to viewpoint changes rather than relying on a fixed observation configuration.

RynnBrain-VLA also performs well on the Unitree G1 humanoid. On *Pull the Chair*, which involves coordinated humanoid whole-body motion, RynnBrain-VLA achieves a 90% success rate, compared with 75% for GR00T N1.7 combined with the same SONIC controller. Since GR00T N1.7 with SONIC is a natural baseline for predicting SONIC motion tokens, this comparison provides additional evidence that RynnBrain-VLA can produce reliable actions even for challenging whole-body control tasks. This SONIC-specific representation is used only for G1 deployment and is distinct from the 81-dimensional unified action space used as the common policy interface for cross-embodiment training. Overall, the results demonstrate that its real-world performance is stable not only across tasks, but also across robot morphologies, viewpoints, and control configurations.

5.6.3 A Stronger Embodied Base Yields a Better VLA.

We next investigate whether an embodied foundation model provides a stronger initialization for VLA post-training. RynnBrain-VLA and Qwen-Based-VLA follow the same VLA post-training recipe and use the

same action-chunk length, while being initialized from RynnBrain and Qwen Based Model, respectively. As shown in Table 6, RynnBrain-VLA consistently outperforms Qwen-Based-VLA across all three long-horizon tasks. Its average process score improves from 68.33% to 90.94%, while the average final success rate increases from 60.00% to 85.00%. The advantage is particularly pronounced on *Grab the Spatulas*, where RynnBrain-VLA achieves a 95% success rate compared with 50% for Qwen-Based-VLA. These results suggest that the capabilities acquired during embodied foundation-model training transfer effectively to downstream VLA post-training. Compared with directly post-training a general-purpose VLM as a VLA policy, initializing from RynnBrain leads to more reliable execution of multi-stage manipulation tasks and substantially fewer failures during intermediate stages. This comparison provides direct evidence that a stronger embodied base model can serve as a better foundation for real-world VLA policies.

5.6.4 Cross-Embodiment Generalization

To examine whether the capability of RynnBrain-VLA is tied to a specific robot embodiment, we adapt it to three morphologically and mechanically distinct robot platforms through the deployment framework described in Section 4.3. As shown in Table 6 and Figure 6, RynnBrain-VLA achieves stable performance across humanoid robots, dexterous-hand manipulation, whole-body control, and bimanual grasping settings. These embodiments differ substantially in morphology, action dimensionality, and low-level control interfaces. The consistent performance across these settings suggests that the embodied representations learned by RynnBrain capture task- and interaction-level knowledge that is not tightly coupled to a particular robot morphology or action parameterization, allowing the same embodied foundation model to serve as a shared starting point for heterogeneous robot policies.

More interestingly, we find that jointly training multiple tasks and embodiments within a single VLA does not introduce obvious interference. With the unified action-space strategy in Section 4.2, RynnBrain-VLA_{Generalist} improves the average process score from 90.94% to 94.74% and the final success rate from 85.00% to 91.67% compared with separately fine-tuned single-task policies. We hypothesize that different embodiments share common interaction structures—such as object affordances, task progress, and reach-grasp-manipulate patterns—even though their executable actions are different. Joint training therefore allows these shared structures to reinforce a common visual-action representation, while unified action-space preserves embodiment-specific control. This suggests that cross-embodiment data can act as complementary supervision: the key benefit of a unified VLA may lie not only in supporting more robots with one model, but also in allowing different embodiments to teach each other how the physical world can be interacted with.

6 Conclusion and Future Work

We presented **RynnBrain 1.1**, an embodied foundation model family spanning 2B, 9B, and 122B-A10B scales. RynnBrain 1.1 studies embodied capability scaling under a unified training framework and demonstrates clear improvements in multi-view reasoning, spatial grounding, and robot-oriented understanding. We further introduce explicit 3D-grounded supervision for the 2B and 9B models, enabling native language-conditioned 3D grounding, together with a compact contact-centered representation for interaction grounding.

We also develop RynnBrain-VLA to connect embodied understanding with real-world action. Under the same post-training recipe, RynnBrain-VLA substantially outperforms a Qwen-based VLA on long-horizon manipulation tasks. Through a unified action space with embodiment-specific masking, the policy is deployed across a Unitree G1 humanoid, an Astribot bimanual robot, and a Tianji-Wuji dexterous-hand system. Joint multi-task and multi-embodiment training further improves real-robot performance, providing initial evidence that heterogeneous robot data can contribute complementary supervision to a shared policy.

Looking forward, we aim to extend RynnBrain toward a unified multimodal model that combines perception, understanding, generation, spatial grounding, and action within a common framework. An important direction is to develop RynnBrain into the cognitive core of a general embodied agent, integrating long-term memory, world modeling, task planning, active perception, and closed-loop interaction. Such an agent could continuously reason about changes in the physical world, generate and revise plans from feedback, and acquire new skills through interaction across diverse tasks and robot embodiments.

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Appendix

A Prompts Details

A.1 Training QA Prompts

To ensure reproducibility, we list the detailed prompt templates used for each training task in [Table 7](#). All prompts are presented in a unified Python f-string format.

A.2 Evaluation and Inference Prompts

To ensure reproducibility, we list the detailed prompt templates used for each benchmark in [Table 8](#). All prompts are presented in a unified Python f-string format.

Table 7 Prompt Templates for Various Training Tasks

Training Task	Prompt Template (Unified Format)
Object Understanding	<code>f"I'd like to know about the area labeled <object> ({x[0]}, {y[0]}), ({x[1]}, {y[1]}) </object> in the image. Can you give a short description?"</code>
Spatial Understanding	<code>f"You are in the last frame of the video. There are {n} objects in the video: <object{idx}> <object> <frame {frame_id}>: ({x[idx][0]}, {y[idx][0]}), ({x[idx][1]}, {y[idx][1]}) </object>, ... \n{question}"</code>
OCR	<code>f"Locate the text \"{text}\" in this video.\n\nStep 1: Predict the key frame.\nStep 2: Output a tuple series.\nOutput format: <area> <frame n>: (x1, y1), (x2, y2), </area>\nwith all coordinates normalized to 0-1000 range."</code>
Object Location	<code>1. f"{question}\nOutput the bounding box in the format <object> <frame n>: ...; (x1,y1), (x2,y2) </object>. n is the chosen frame index." 2. f"{question}\nGenerate coordinates for one object bounding box. Constraints: x1,y1,x2,y2 ∈ [0,1000]. Response must be in the format: <object> (x1, y1), (x2, y2) </object>"</code>
Area Location	<code>1. f"{question}\nFirst perform key frame prediction, then generate a sequence of coordinate tuples.\nOutput format: <area> <frame n>: ...; (x1, y1), (x2, y2), </area>\nEach coordinate pair must contain normalized pixel values within the [0, 1000] range." 2. f"{question}\nExpress the coordinates as a tuple sequence in the format <area> (x1, y1), (x2, y2), ... </area> with all coordinate values normalized to the standardized pixel coordinate system spanning 0 to 1000."</code>
Affordance Location	<code>1. f"{question}\n1. First identify the key frame\n2. Then predict one affordance point.\nOutput format: <affordance> <frame n>: ...; (x, y) </affordance>\nCoordinates normalized to 0-1000 pixel space." 2. f"{question}\nTask: Affordance point prediction\n- Identify one possible affordance point\n- Normalize coordinates to 0-1000 range\n- Output format: <affordance> (x, y) </affordance>\n- Example: [450, 320]"</code>
Trajectory Location	<code>1. f"{question}\nor trajectory completion:\n1. First locate the frame with the trajectory start point\n2. Then predict up to 10 key points as list. Output format: <trajectory> <frame n>: ...; (x1, y1), (x2, y2), </trajectory>\nAll coordinates normalized to 0-1000 pixel space." 2. f"{question}\nPredict a trajectory comprising up to 10 key points. Return coordinates in the format <trajectory> (x1, y1), (x2, y2), ... </trajectory> with all values normalized to the [0, 1000] range."</code>

Table 8 Prompt Templates for Various Benchmarks

Benchmark	Prompt Template (Unified Format)
VSI-Bench	1. <code>f"{question}\nAnswer with the option's letter from the given choices directly."</code> 2. <code>f"{question}\n\nAnswer the question with an exact number, which should be accurate to at most two decimal places."</code>
MMSI	<code>f"{question}"</code>
ERQA	<code>f"{question}\nAnswer with the option letter from the given choices directly."</code>
RoboSpatial	1. <code>f"{question}. Pinpoint several points within the vacant space situated to the left of the vacuum. Your answer should be formatted as a list of tuples, i.e. [(x1, y1), ...], where each tuple contains the x and y coordinates... indicating the normalized pixel locations of the points."</code> 2. <code>f"{question} Answer yes or no."</code>
MindCube	<code>f"Select the best answer to the following multiple-choice question based on the image.\n{question}\nAnswer with the option's letter from the given choices directly and only give the best option. The best answer is: "</code>
RynnBrain-Object	<code>f"{question} Your current position is at the last frame of the video."</code>
RynnBrain-Spatial	<code>f"{question}"</code>
RefSpatial-Bench	<code>f"Locate {object_name} in this image. Output the point coordinates in JSON format."</code>
RynnBrain-Grounding	<code>f"{question}. Output the bounding box in the format <object> <frame n>: ...; (x1,y1), (x2,y2) </object>. n is the chosen frame index."</code>
RynnBrain-Area	<code>f"{question}. First predict the key frame, then output coordinates as a series of tuples. \nOutput format: <area> <frame n>: ...; (x1, y1), (x2, y2), </area>\n All coordinates must be normalized between 0 and 1000."</code>
RynnBrain-Affordance	<code>f"{question}. First predict the key frame, then output a single affordance point as coordinates (x, y).\nOutput format: <affordance> <frame n>: ...; (x, y) </affordance>\n Both x and y values must be normalized between 0 and 1000."</code>
RynnBrain-Trajectory	<code>f"{question}. First predict the frame containing the trajectory start point, then output up to 10 key trajectory points as a list of tuples in the format: <trajectory> <frame n>: ...; (x1, y1), (x2, y2), </trajectory> All coordinates must be normalized between 0 and 1000."</code>

A.3 Hyper-parameters for Evaluation

To ensure reproducibility, we disabled sampling during autoregressive text generation unless otherwise specified. For images processing, we constrained the resolution by setting `min_pixels` to $16 \times 32 \times 32$ and `max_pixels` to $16384 \times 32 \times 32$. For video-related benchmarks, frames are sampled at 2 FPS; if the total exceeds 512 frames, we apply uniform sampling to maintain a maximum of 512. The `min_pixels` for each frame and `max_pixels` for the whole video are set to $16 \times 32 \times 32$ and $24576 \times 32 \times 32$, respectively. On certain pointing-related benchmarks (e.g., ERQA, RoboSpatial, and ShareRobot), we observed that sampling can further enhance model performance. In these instances, we set the temperature to 0.2, `top_p` to 0.95, and `top_k` to 50. Additionally, for RefSpatial and ShareRobot—which involve numerous precise positioning tasks—we employed a higher resolution by increasing `min_pixels` to $1024 \times 32 \times 32$.